Chapter 5
Functional properties of food

Have you ever wondered how so many different and interesting foods can be made using the same basic ingredients? Combine flour, butter, eggs, milk and sugar in different ways and you could end up with Portuguese tarts, soufflés, brownies or bread. What happens to food when it is prepared or cooked is controlled by its functional properties. Understanding the functional properties of food will enable you to interpret recipes and assist you in creating better quality foods. Knowing why is as important in food preparation as knowing how!

In this chapter you will learn about:

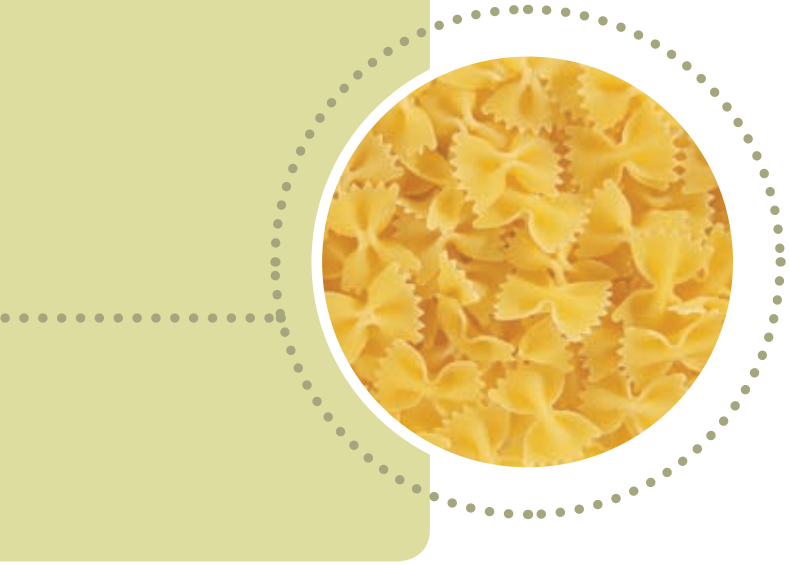
- the role of proteins in denaturation, coagulation, gelation, foaming and browning
- the role of carbohydrates in gelatinising, dextrinising, caramelising and crystallising
- the role of fats in emulsifying and aerating.

You will also learn about the factors that can affect the functional properties of food, including:

- oxygen
- temperature
- acidity
- agitation
- enzymes
- addition of other ingredients.



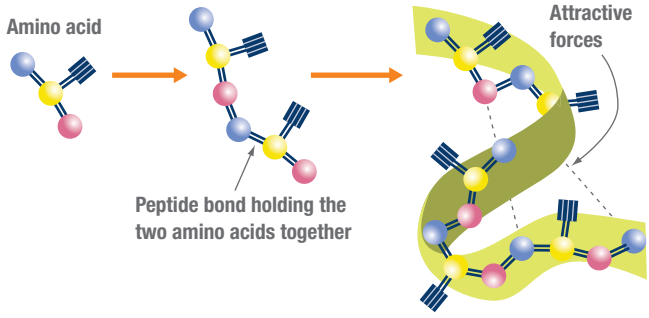
Making food look this good is a skill. As with all skills, practice makes perfect. Understanding what is happening to the ingredients as they are prepared increases the chance of creating a masterpiece. Do you ever go to a specific shop and buy a particular food because they make it so well there? Try analysing the characteristics of the food, and use the information in this chapter to assess why another shop's product is not as good.



Given that nutrients determine the way a food looks, tastes and feels, this chapter investigates the main nutrients and the changes that each one undergoes.

Proteins

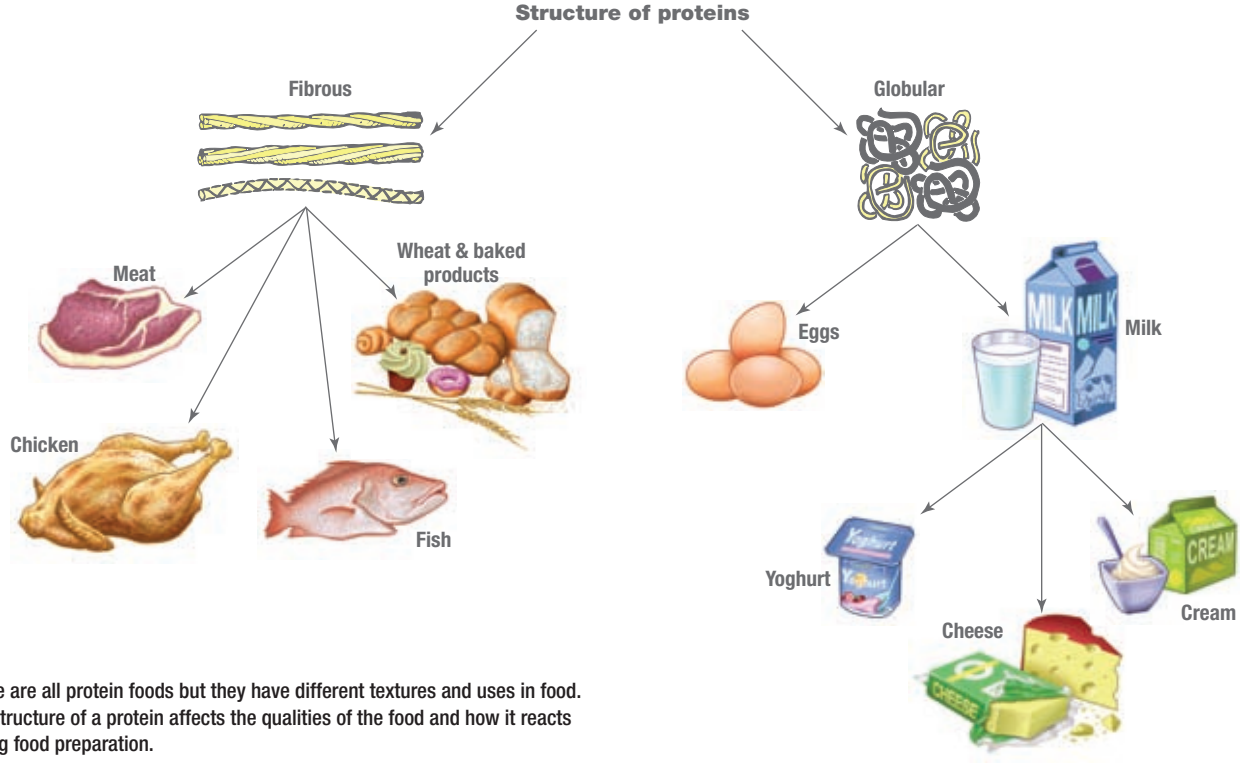
We all know that **protein** is essential to our diet and can be obtained from foods such as meat, fish, dairy products and eggs. But, scientifically, what is a protein? Proteins are made up of individual building blocks called **amino acids**. There are 22 amino acids, which can be combined in many ways to produce thousands of different proteins. Our bodies contain about 30 000 different proteins, of which only 2 per cent have been accurately described. Individual amino acids are joined together by **peptide bonds** to form polypeptide chains, much like a single strand of beads. Each strand twists back on itself to form a coil or spring shape because one part of the amino acid attracts another. These attractions usually occur between hydrogen atoms or wherever sulfur occurs as part of an amino acid.



The polypeptide structure of a protein is coiled because attractive forces exist between one amino acid and another.

The shape of a protein chain determines the characteristics of the foods in which it is found. Which do you think would be stronger: a protein with a twisted strand structure or one with a rounded structure? Which structure do you think would be the easier to alter during food preparation?

Strands of protein always arrange themselves in groups of three or four in a coiled or **helix** shape. Attractive forces not only operate along a single protein strand, causing it to coil, but also exist between protein strands. If the sulfur or hydrogen atoms in one protein chain are positioned close to the equivalent in another protein chain, they either react to form a disulfide bridge (often called a salt bridge) or are attracted and form a hydrogen bond (or H bond).



These are all protein foods but they have different textures and uses in food. The structure of a protein affects the qualities of the food and how it reacts during food preparation.

These attractions and reactions determine the shape of the protein, which can be in one of two basic structural forms:

1. fibrous proteins, where the coiled helix structure is elongated to form a strand. Fibrous proteins are found in the muscle fibres of meat, poultry and fish, and also in products made from flour that contain the protein gluten.
2. globular proteins, where the protein stands are twisted into a rounded, compact shape. Globular proteins are found in:

- eggs — albumin in eggwhite, and vitellin and vitellenin in egg yolk
- milk and dairy products — caseinogen, lactalbumin and lactoglobulin
- enzymes — present in the cells of all living things. The presence of enzymes in foods is most easily seen in fruits and vegetables; they are responsible for the process of ripening and for the browning reaction that takes place when fruits are cut and exposed to air (**enzymic browning**). Enzymes are also used by food processors, such as cheesemakers, to create specific reactions in foods.

What happens to proteins during food preparation and processing?

Each protein has its own combination of amino acids — its own fingerprint. This means each protein is slightly different from all other proteins, has specific functions and undergoes specific reactions. However, there are some reactions common to *all* proteins, the two most important in food being **denaturation** and **coagulation**. These changes occur when the structure of a protein is altered by:

- the application of heat
- mechanical action (agitation)
- the presence of acids or salt
- enzymic action.

Denaturation occurs when the bonds holding the helix shape are broken and the strands of the helix separate and unravel. Whether this change is visible in food depends on the protein type. Globular proteins denature very easily and denaturation is easier to see. Add some lemon juice to a glass of milk and observe the change in the texture of the milk. The foam created when beating eggwhites is also a visible sign of denaturation. Acidophilus (acid-producing) bacteria are present in yoghurt for more than just their nutritional benefits; they denature milk protein to make it more digestible.

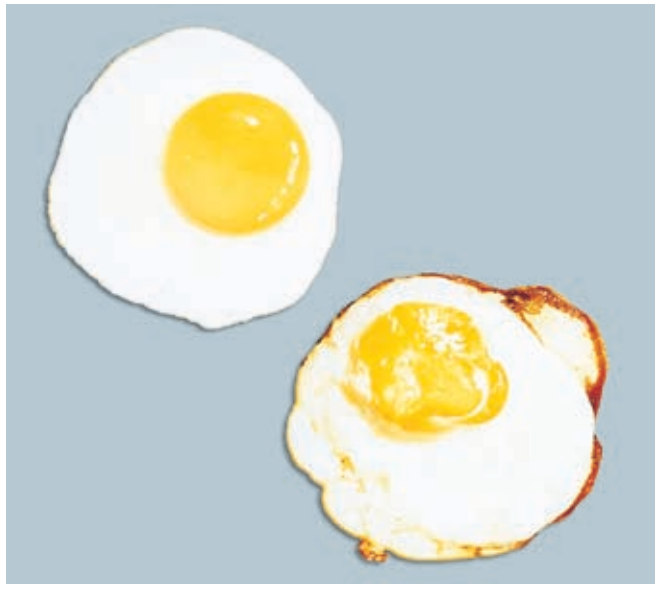
The denaturation of fibrous proteins, however, is less visible. The structure of the protein is stronger and therefore less affected by factors that easily alter the structure of globular proteins. Given time, however, fibrous proteins denature and the process is identifiable. For example, tough cuts of meat are often marinaded overnight in acid ingredients to make them more tender. The acids denature the collagen, reducing its strength and reducing the amount of pressure needed to break it apart, thereby giving the meat a more tender mouth feel.

Coagulation is a more visible process than denaturation; it occurs when denatured proteins separate from other nutrients and solidify. When denaturation continues to the point that the unravelled protein strands begin to rejoin with other strands, a solid mass is formed. These coagulated protein strands can bind liquids to form soft, elastic solids **(gels)**; for example, mixtures containing gelatin thicken when they cool. However, if proteins overcoagulate, they shrink and toughen, forcing liquid out from within the proteins.

Factors contributing to denaturation and coagulation

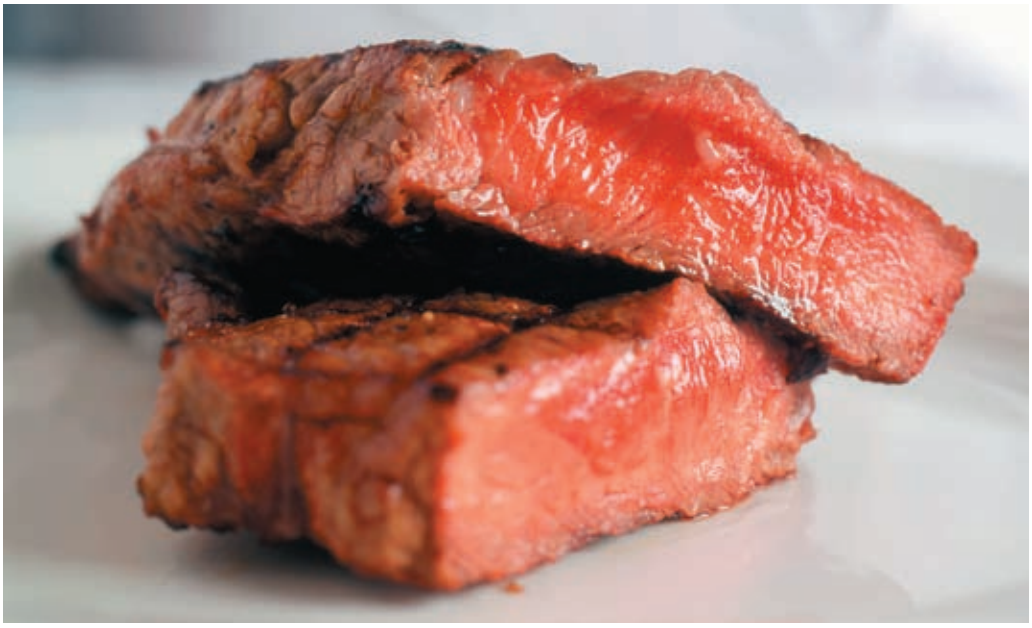
Temperature

Heat causes proteins to denature or unravel. The higher the temperature is, the more quickly proteins denature and then coagulate or clot. Even the temperatures during preparation can affect the functional properties of proteins. For example, beating eggwhites that are at room temperature gives a different volume of foam than eggs taken straight from the fridge. Which do you think would give you more foam?



The difference between an enjoyable breakfast and a disappointing one! Time and temperature control are important to maintain the quality of proteins. When an egg cooks, the clear eggwhite (which is mainly protein) starts to coagulate at 60 °C and the yolk is firm when the temperature reaches 68 °C. If you continue to cook the egg, the protein shrinks.

Cooking meat also illustrates the effect of temperature on proteins. Whenever you order steak at a restaurant and are asked how you would like it cooked, you are actually advising the chef whether you would like all the proteins in the steak coagulated (a 'well done' steak) or partially denatured and coagulated (such as 'medium rare'). As the internal temperature of the meat increases during cooking, so does the degree of coagulation.



How would you order your steak at a restaurant? Would you order it slightly or fully coagulated? What is it about the texture of the meat that makes you prefer it cooked that way?

EXPERIMENT

Does low temperature affect the functional properties of proteins?

Syllabus outcome

Students learn to:

- investigate through experimentation the factors that affect the functional properties of foods.

Contributes to the following outcome:

- apply an understanding of the sensory characteristics and functional properties of food to the preparation of food products.

Aim

This activity seeks to show the impact of low temperature on the functional properties of proteins. When beaten, eggwhites can denature and form a foam. Eggwhites that are cool when beaten produce a different quantity of foam from eggwhites beaten when at room temperature.

Equipment

For this experiment, some groups will use refrigerated eggs while other groups will use eggs at room temperature. All other ingredients must be refrigerated. The following equipment should be chilled in the freezer for at least 15 minutes before use:

- 2 medium bowls, metal if possible
- 1 or 2 sets of beaters from electric mixers
- metal spoon or rubber spatula
- shallow metal trays (2 bar tins or 1 lamington tin)
- aluminium foil

Groups using eggs at room temperature need to chill only one of the two bowls.

Method

All factors relating to the production of the ice-cream should be the same, including the same size and shape of equipment. The quality of the ice-cream depends on the speed of preparation; groups must be organised and work quickly so that their ingredients remain cold.

Making ice-cream

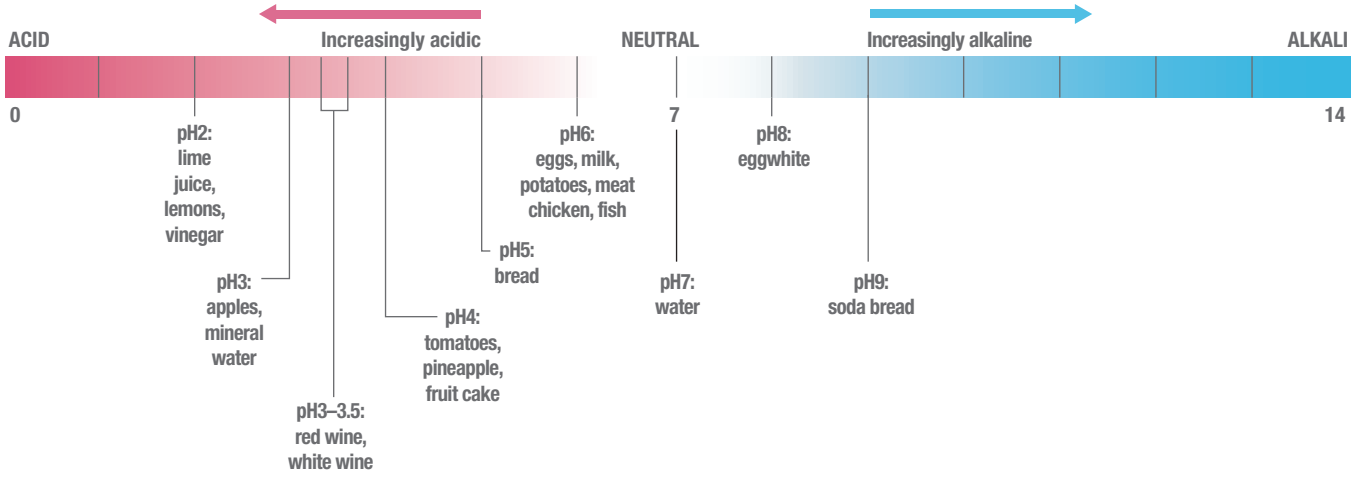
1. Separate the egg yolks from the eggwhites. Put the eggwhites into another bowl.
2. Whip the cream until stiff peaks form. Add the egg yolks and vanilla, beating slowly until lightly combined.
3. Fold flavourings into cream using spatula or metal spoon. Refrigerate while beating eggwhites.
4. Beat eggwhites to soft peaks. Gradually add sugar while beating, and continue until stiff peaks form.
5. Fold eggwhites into cream mixture.
6. Remove metal trays from the freezer, fill and cover with foil.
7. Put ice-cream in the coldest part of the freezer and allow to freeze overnight.

Ice-cream recipe

2 large eggs or 3 small eggs
300 mL fresh cream
3 teaspoons vanilla
⅓ cup of one or more flavourings: finely chopped or grated chocolate, chopped cherries, mango purée, frozen berries, syrup or topping, pecans, sultanas (hydrated in fruit juice)
½ cup caster sugar

Results and conclusions

1. Before making the ice-cream, predict which of the ice-creams will have the greater volume. Give a reason for your prediction.
2. Compare the volumes of the ice-cream before being put in the freezer. Were your predictions correct? If not, what reasons can you suggest for the different volumes?
3. Compare the volumes of the ice-cream after freezing. Suggest reasons why the volume may have decreased during freezing.



The pH scale of common foods and liquids

Acidity

When the pH is less than 7 (indicating acidity), proteins will denature. **pH** refers to the acidity or alkalinity of a substance and is measured on a scale of 1–14. A substance with a pH of 7 is considered to be neutral — neither acid nor alkaline.

Acids cause proteins to denature by attacking and breaking the salt bridges between amino acid strands. We use this to our advantage when we marinade meats in acidic marinades or create a range of thickened dairy products. Sour cream and yoghurt are both made by adding lactic acid-producing **bacteria**, in a **bacterial culture**, to pasteurised milk. During processing, the bacteria produce lactic acid, which causes the protein caseinogen in milk to coagulate and the product to develop a slightly sour taste. It should be noted, however, that food manufacturers are simply recreating a natural process in a controlled environment. Milk has always contained these bacteria; they are responsible for the souring of milk and the accompanying physical changes. The curdling of soured milk is simply coagulation and the odour is a by-product of the bacteria present. The souring of milk occurs more quickly at temperatures above 5 °C as bacteria are more active in warm environments.

Acids have a sour taste and a pH reading below 7. Common acids found naturally in foods include citric acid (from citrus fruits such as lemon and limes) and acetic acid (found in vinegar). All of these acids are considered weak acids because their pH is above 2.0. Lemons and limes have a pH of 2.0–2.2. (See table 5.1 for common acids in foods.)

Alkalis have a bitter taste, feel slippery in the mouth and have a pH above 7. Alkalis also denature proteins, but they are not commonly found in food, except for sodium bicarbonate (baking soda).

Agitation

Agitation causes proteins to denature. Agitation in food handling is usually in the form of kneading, mixing, whipping or beating. Such mechanical action during food handling causes the protein strands to stretch. This is clearly desirable when making ice-cream, bread and most cakes.

TABLE 5.1 Some common acids found in foods

ACID	FOOD SOURCES
Citric acid	Citrus fruits (oranges, grapefruits, lemons, limes), cranberries, raspberries
Malic acid	Apples, peaches, grapes, rhubarb, cherries, apricots, strawberries
Oxalic acid	Rhubarb, ripe pineapples
Acetic acid	Vinegar
Tartaric acid	Grapes, baking powders
Benzoic acid	Cranberries

Acids are essential ingredients in many foods and they contribute to the specific characteristics of the food. Acids may also be added by manufacturers to give specific characteristics. An additive number code listed on a label does not mean the additive is artificial; it may just mean that it does not occur naturally in the food in the quantities needed to provide the desired characteristics. Brainstorm some reasons why manufacturers may add acids to foods.

However, too much mixing or beating stretches the strands until they break and denatures the protein. Overbeating eggwhites causes them to clot, which is not acceptable for products requiring a smooth, fluffy texture such as a pavlova.

Enzymes

Enzymes are chemical substances that act as catalysts in chemical reactions. **Catalysts** assist chemical changes without actually becoming involved in the reaction.

Human digestion would not be possible without enzymes. Enzymes make it possible for the body to break down large molecules, such as protein molecules, into smaller units so they can be absorbed and used by the cells. The protein-splitting enzymes that assist with the digestion of protein break the strands of amino acids apart. The destruction of the peptide bonds then breaks each strand into its individual amino acids.

It is also possible to use enzymes when preparing food. Enzymes can be used to tenderise meat. Enzymes found naturally in the meat can denature the tough protein elastin, but this **ageing process** takes a number of days. Soaking meat in a marinade containing one or more protein-splitting enzymes can speed up denaturation. A marinade containing pineapple juice, for example, contains the enzyme bromelin. The bromelin denatures the protein myosin on the outside of the meat muscle, as well as the elastin and collagen in connective tissue surrounding the muscle. If a fork is used to pierce the meat, the enzyme can act on inner layers as well. Denaturation speeds up when heat is added, so marinades work better at room temperature than in the refrigerator. Two other common tenderisers are papain (made from unripe pawpaw) and ficin (from figs).

The enzyme rennin is sometimes used to coagulate milk in cheesemaking. Rennin, which occurs naturally in the stomachs of calves and human infants, is commercially available in the form of a powder called rennet. The function of rennin in the body is to clot milk for digestion into a soft solid coagulated mass, called a curd. In cheesemaking, the clotted mass is cut with fine wires to break the curd, releasing a liquid called whey. The curd is then processed further with heat, agitation, salt and other ingredients to produce a variety of cheeses. The whey is heated again to coagulate any remaining proteins, producing ricotta cheese.



Have you ever thought of making your own cottage cheese or yoghurt? These foods have been made traditionally by hand for centuries. Why not give it a go? You can buy junket powder and yoghurt starter kits from the supermarket. Research some recipe ideas in cookbooks or on the internet.

Salt

Sodium chloride or salt can denature and coagulate proteins. Have you ever wondered why recipes for boiled eggs state they should be cooked in salted water? If the eggshell cracks during cooking, the salt in the water helps minimise the loss of white from the shell by increasing its rate of coagulation and sealing the hole with solid eggwhite. Salt is added when making cheese to increase the firmness of the curd; it does this by increasing coagulation of the curd and forcing water out from within the proteins. Salt also helps protect the cheese from the growth of micro-organisms.

CASE STUDY

Making cheddar cheese



1. Raw milk entering the factory is tested for quality, including microbial levels, chemical residues (medications given to cows) and cream content.
2. The milk is pasteurised (heat treated) to destroy harmful bacteria before being cooled and poured into large containers called vats, each holding around 20 000 litres.
3. Lactic-acid-producing bacteria and rennet are added to the milk to coagulate the proteins. The bacteria also contribute to the flavour of the cheese, while the rennet is responsible for setting the milk as junket (thickened protein curd holding liquid whey). The mixture is warmed slightly and agitated so that coagulation occurs quickly.
4. The curd is cut using wires to release the whey. The mixture is then agitated gently and heated to 37 °C for around 2 hours to allow the curd to shrink and become denser as more whey is forced out from the proteins.
5. The whey is drained from the curd and the small particles of curd are fused with pressure and warmth into blocks. The blocks are turned every 10 minutes for an hour to ensure removal of the whey and the development of an even consistency within the cheese. During this process (called cheddaring), the lactic acid level within the cheese increases, providing cheese with its characteristic flavour.

6. The cheese curd is then milled, salted and mellowed. During milling, the cheese is cut into small lengths resembling French fries. This ensures that the salt added in the salting process has access to all parts of the curd. Once salted, the cheese curd is rested, or mellowed, for about 20 minutes to allow the salt to penetrate the curd.
7. The cheese curd is then formed into large blocks weighing about 20 kilograms. This is done by placing the curd into a block-forming tower where it is compressed, using its own weight and a vacuum. At the bottom of the tower, the blocks are vacuum-sealed in plastic, weighed and labelled, and then boxed for storage.
8. The cheese is cooled and stored at 8 °C for specific lengths of time, depending on the type of cheese. This process is called maturing. Mild cheeses are matured for 3 months, tasty cheeses for up to 9 months, and vintage cheeses for 15 to 24 months.

OUTCOME TASK

Students learn to:
- explain some of the functional properties of food
- identify the factors that affect the functional properties of food.

Contributes to the following outcome:
- identifies and explains the sensory characteristics and functional properties of food.

Carefully consider the case study above. Remember that this is not just a comprehension task; use the stimulus material along with your knowledge of *the factors affecting the functional properties of food* to complete the tasks ahead.

1. In the first stage of processing, the milk is heated (pasteurised) to destroy bacteria. Suggest why this might be done when the process of cheesemaking then requires bacteria to be added to the milk.
2. Explain the effects that lactic-acid-producing bacteria and rennet have on the milk. How would adding both to the milk affect the consistency of the cheese curd?
3. Why would agitation of the milk be important during the heating process?
4. From your knowledge of bacteria and enzymes, explain why the milk is heated to 37 °C and not 65 °C.
5. When the curd reaches the mellowing stage, it is still soft and moist. Describe how the salt affects the consistency of the curd during mellowing, giving factual reasons to support your answer.

The influence of sugar on protein structure

If sugar is near protein when heat is added, the protein can tolerate higher temperatures before denaturing and coagulating; in other words, sugar protects the protein structure. In custard, for example, the presence of sugar means that more heat is needed to coagulate the protein in the milk and eggs to thicken the custard. A runny custard means that not all the protein has coagulated. For the same reason, a custard gel with a high sugar content may not be very stable and may weep.

Another example of sugar protecting the protein structure is the honey glaze on baked ham, which prevents the outer layer from becoming tough.

Formation of protein gels

Gelation is the process of gel formation. You may recall that gels have been described as elastic solids. Gelation can occur when protein or carbohydrate reacts in certain conditions. We will discuss the gelation of proteins here; the gelation of carbohydrates will be discussed later in this chapter.

Gelation of proteins occurs when tiny droplets of liquid are trapped within coagulated proteins, forming soft, elastic products such as custard, junket and cheese. The presence of heat, acids or enzymes are necessary for the formation of these gels. Once coagulation has occurred, gelation occurs more quickly if both the pH and temperature of the mixture are low.



It's easy to make jelly! Empty contents of packet into a 500 mL jug and add 1 cup boiling water. Stir and dissolve well. Add 200 mL cold water and stir. Pour into bowl. Refrigerate until firm (minimum 4 hours). Makes 500 mL of jelly. Do not add fresh pineapple, kiwifruit or pawpaw or jelly will not set.

The ingredients found in jelly of this type could include sugar, gelatin, food acids (297, 331), flavour and colours (in this packet it is 122, 133). Can you explain why each of these ingredients would have been added? Can you explain the reasons for each step of the recipe? If you're unsure, revise the concepts on the previous pages.

When most people think of food gels they think of jelly. Do you realise that the gelatin used to create jelly is actually a protein? Gelatin is a globular protein that denatures and unravels in the presence of moisture and heat. As the temperature drops, the strands of gelatin trap the surrounding liquids, forming a gel. If the pH of the gel is around 2.8, the gel sets at room temperature.

Summary

In summary, the causes of protein denaturation and coagulation are:

- the application of heat — for example, the toughening of meat as it cooks
- an acidic pH — for example, when lactic acid is added to milk in cheesemaking
- agitation — for example, whipped eggwhites
- enzymes — for example, marinaded meat
- salt — for example, when poaching eggs.

The texture of a coagulated protein product is influenced by:

- the speed of coagulation. For example, cooking meat too quickly causes the proteins to overcoagulate and become dry and tough.
- the presence of sugar. Sugar supports and stabilises protein gels in small quantities.
- the presence of fat. Fat surrounds proteins such as gluten and ovomucin and limits their interaction with other proteins.

REVIEW QUESTIONS

Remember

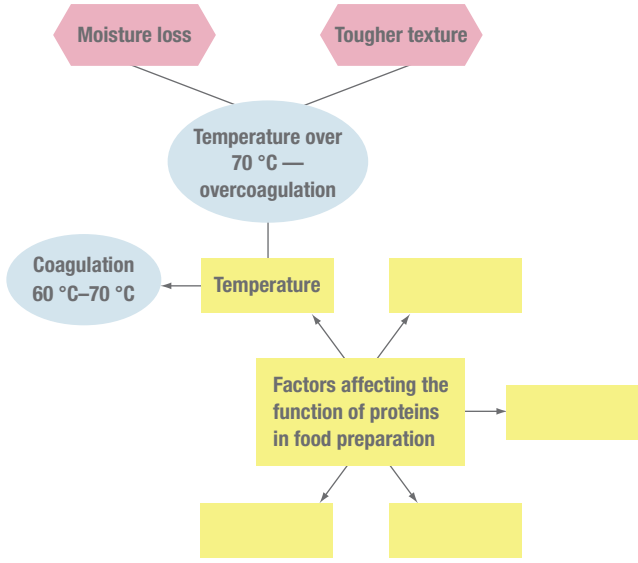
1. Outline the two different structures of proteins and give at least two food sources of each.
2. Identify the five factors that affect the physical qualities and functional properties of protein foods.

Apply

3. Describe how the coagulation of proteins differs from denaturation.
4. Identify both the functional properties used in the preparation of the following foods and the factors that affect them. (*Note:* More than one functional property or factor may contribute to the characteristics of the product.)
   a) Pavlova and meringue
   b) Fetta cheese
   c) Marinated meat
   d) Grilled meat
   e) Baked custard

Do an activity

5. Using the following diagram as a starting point, create a concept map identifying the different factors that affect the functional properties of proteins, include examples of how each property is affected. Compare your concept map with your classmates, and then work together to create a concept map poster for display in the classroom.



6. Log in to **www.jacplus.com.au** and locate the *Make your own cottage cheese* weblink for this chapter. Make some of your own cheese and apply your understanding of the functional properties of proteins as you do.

eBookplus
Weblink

Functional properties of specific proteins

We have just examined the two properties that all proteins possess — the ability to denature and coagulate. Now we will examine specific proteins to identify the additional function(s) that each can carry out during food preparation and processing.

Aeration of eggwhites

The process of adding air to a substance is called **aeration**. When eggwhites are beaten to make meringues, soufflés and certain cakes, air is mixed into the protein of the eggwhite, creating a foam.

The protein ovomucin in eggwhite separates from the rest of the eggwhite when it is beaten. As it denatures, it coils and traps air, creating large bubbles. Continued beating causes the bubbles to become smaller and the foam becomes whiter and finer in texture. The film of liquid protein around the bubbles becomes thinner, while the proteins within the film become more solid. Eventually, the foam holds its shape when the beaters are removed, forming stiff white peaks. How the foam is handled now determines the quality of the food item produced:

- If the foam is beaten for too long, the denatured protein coagulates, the surface shrinks, and liquid weeps from the clumps of protein.
- The foam must be used immediately. For example, it could be spread on a baking sheet or piped onto a pie, and put into the oven. The foam is fragile and, after a few minutes,

the stretched protein loses elasticity, the air bubbles collapse under the weight of the foam, and liquid collects at the base of the mixture.

- If the foam is heated, the trapped air rises, and the protein stretches to give structure to the food. The protein coagulates between 60 and 65 °C and the food takes on its permanent shape.

The aeration of eggwhites is affected by the following factors, which change the functional properties of nutrients.

1. *Temperature*. Eggwhites aerate more rapidly and produce foams of a higher volume at room temperature. Refrigerated eggwhites are less elastic and produce smaller and fewer air bubbles, resulting in a smaller volume of foam.
2. *Acids*. Small amounts of acid, such as lemon juice and cream of tartar, may slow the formation of foam but increase its stability, giving slightly more time to handle the foam before cooking.
3. *Agitation*. The greater the rate of agitation, the greater the volume of foam produced. Also, if finer blades, like those of a wire whisk, are used, the volume of foam is greater and the texture of the foam finer.
4. Addition of other ingredients:
   - Fat (even small amounts of yolk in a bowl or on beaters) prevents eggwhites forming a foam. For this reason, ensure that the bowl and beaters are clean and free from grease.
   - Water (up to 40 per cent of the volume) increases the volume of beaten eggwhites but makes them less stable.
   - Sugar added towards the end of beating makes the foam shinier, finer in texture, more stable and, therefore, less likely to weep.



Globular proteins visibly denature and partially coagulate when aerated to a foam. Specific conditions are required for foams to occur. Complete the following experiment to examine which conditions assist in producing a stable eggwhite foam, and which conditions reduce this functional property.

EXPERIMENT

The effect of various additives on eggwhite foams

Syllabus outcome

Students learn to:

- investigate through experimentation the factors that affect the functional properties of foods.

Aim

To determine the effects of salt, cream of tartar, oil, lemon juice and water on the volume, beating time and colour of beaten eggwhites

Equipment

5 eggwhites
5 small bowls of same size
electric mixer
¼ teaspoon salt
¼ teaspoon cream of tartar
¼ teaspoon lemon juice
¼ teaspoon oil
2 teaspoons water

Method

Divide the class into five groups. Each group observes the effects of one of the additives on eggwhite.

Group 1 — ¼ teaspoon salt
Group 2 — ¼ teaspoon cream of tartar
Group 3 — ¼ teaspoon lemon juice
Group 4 — ¼ teaspoon oil
Group 5 — 2 teaspoons water

1. Put one eggwhite in a small bowl. Record the time.
2. Beat on medium speed until foamy. Record the time.
3. Place the additive in the foam. Record the time.
4. Start beating again and beat until stiff peaks form.
5. Compare the appearance, volume and texture of each whipped eggwhite. Record your observations in a table like the one at the top of page 91.

Conclusions

Which additives:

a) increased the time needed to beat the eggwhites to the stiff-peak stage
b) increased the overall volume
c) altered the colour of the eggwhite?

Results

ADDITIVE	TIME						
	START	SOFT PEAKS	STIFF PEAKS	COLOUR	LUSTRE	VOLUME	SMOOTHNESS OF SURFACE
Salt							
Cream of tartar							
Lemon juice							
Oil							
Water							

Gluten

Gluten is a type of protein found in wheat, and it gives flour products their structure. Gluten is actually made from the two proteins glutenin and gliadin. When moistened and mixed with a small amount of fat, glutenin and gliadin stretch to form gluten strands; these gluten strands stretch and become elastic.

When making bread, for example, your kneading motion gently stretches and folds the protein, fat and water mass. This causes the protein molecules glutenin and gliadin to slide past one another. Physical and chemical cross-linking of the amino acid strands takes place. The two proteins join with the fat to form gluten strands.

In the oven, the air trapped inside the gluten strands rises, while the water turns to steam and also rises. The elastic dough stretches and the bread expands upwards in the tin. This upward motion continues until the outside layer of the dough reaches the temperature at which protein denatures, coagulates and hardens, and a crust forms.

If a flour product is removed from the oven before all the gluten solidifies, the product will collapse. This explains why cakes sink in the middle if they are taken out of the oven before they are cooked. The middle of the cake is the last part to cook, so this is where to test whether the cooking process is complete.

The actual amount of gluten protein in a product depends on the class of wheat used. The five classes of wheat are shown in table 5.2.

TABLE 5.2 Protein content of various types of wheat

TYPE OF WHEAT	AMOUNT OF PROTEIN	USES
Prime hard	Highest protein content (13–15%)	Bread, Chinese noodles
Hard	Slightly lower protein content	Pita bread and flat breads such as Lebanese bread
Australian durum	High protein content	Pasta products
Australian standard white (the most commonly grown wheat)	Average protein content	Bread, cakes, confectionery
Soft (produces a fine crumb)	Lowest protein content (10%)	Cakes, pastries, biscuits

The actual protein content of flour is only one of the factors affecting the development of gluten in a flour product. Agitation and the presence of other ingredients also affect the functional properties of this protein.

- Agitation encourages the development of gluten strands. The more a flour mixture is kneaded or stirred, the stronger and less elastic the gluten strands become.
- Presence of fats such as butter can prevent water coming in contact with the flour, thus preventing the glutenin and gliadin from joining to form gluten and developing into elastic strands. When a recipe requires that butter be 'rubbed in' to flour, the process coats the flour particles to 'shorten' the gluten strands created during preparation, making the finished product more crumbly. If large amounts of fat are used, such as in puff pastry, the gluten strands will be so short that the product will 'melt in the mouth'. The fat forms an impervious layer between the layers of gluten and the air trapped during pastry making lifts the layers when it heats and rises.
- Sugar in baked products, such as cakes and brownies, gives them a softer, more crumbly texture. Sugar is highly absorbent and competes with the flour for water, thereby limiting the amount available for the formation of gluten.



Gluten strands give baked products their structure. Gluten is strong and elastic, stretching as the air and steam inside a mixture expand during baking. When gluten reaches 60 °C, it begins to denature and then coagulate, setting its structure and maintaining the new shape of the baked food.

Protein browning

Foods containing protein turn brown during cooking for three reasons — oxidation, application of heat and the Maillard reaction (explained on page 92).

Oxidative browning of red meat

Myoglobin and haemoglobin are the proteins that give red meat its bright red colour. Myoglobin also causes colour

changes in meat. When myoglobin is exposed to the air (for example, when meat is left uncovered in the refrigerator or cold room), oxygen changes the iron present in myoglobin, and a new substance called oxymyoglobin is formed. The meat remains red for a short time, but the oxymyoglobin on the surface of the meat gradually turns into metmyoglobin, which is brown in colour.

myoglobin (bright red) ⟶ oxymyoglobin (red) ⟶ metmyoglobin (brown)

Despite the brown outer surface, the inside would still be red.

High temperatures, freezing temperatures, ultraviolet light, certain metals and the presence of bacteria speed up this oxidative browning. Many people in Australia will not buy red meat that has turned even slightly brown because the meat has a stronger flavour when cooked. Supermarkets in Australia reduce the price of slightly brown meat, while people in other countries are prepared to pay more for it.

To reduce oxidative browning, carefully wrap meat or remove oxygen from the package.

Non-enzymic browning of red meat

During non-enzymic browning, heat must be applied. (Oxygen is not involved as part of the reaction.) When the blood protein myoglobin in meat is heated, it denatures, changing the colour from pink to brown to grey. Because the outside of the meat experiences the heat first, the colour change happens from the outside to the inside. Thus meat can be quite brown on the outside (even black if the heat is high enough, as can happen on a barbecue) and pink on the inside. The colour change is complete when the internal temperature of the meat is 82 °C. At this temperature, all pinkness in the meat is gone, and the meat is described as being 'well done'.

Maillard reaction

The **Maillard reaction** takes place when dry heat is applied to a food that contains both a protein and a sugar. The formation of a dark yellow to brown crust on cooked food — such as roast meat, cakes, bread, custard, fried chips and ready-to-eat cereals — is a result of the Maillard reaction, and it is part of the overall appeal of these foods.

The type of heat is important. Dry-heat cooking methods include baking, roasting, barbecuing, grilling and frying. Moist-heat methods of cooking (such as steaming, boiling and casseroling) and microwave cooking do not produce the Maillard reaction.

The Maillard reaction also occurs when it is unwanted. The process of drying milk or eggs, for example, can produce the reaction; the product becomes slightly brown in colour and has an 'off' taste. If stored food containing both protein and sugar is allowed to heat above normal room temperature, the Maillard reaction can again cause undesirable colour changes. The higher the storage temperature is, the greater the reaction will be.

REVIEW QUESTIONS

Remember

1. Define 'denaturation' and 'coagulation'.
2. **a)** State whether the proteins in eggwhites are globular or fibrous.
   b) What is gluten? Is it a globular or fibrous protein?
3. Why does meat brown when it is cooked?

Apply

4. Aeration is a specific functional property of eggwhites. What factors affect the ability of the eggwhite proteins to form and hold a foam?
5. Apply your knowledge of the functional properties of gluten to explain the following scenarios.
   a) Scone dough is kneaded for 3 minutes when the recipe said to knead it lightly. The resulting scones are dense and flat.
   b) A cake is placed in the oven at too high a temperature. When removed from the oven, its surface is cracked.
   c) When a cake is taken out of the oven, it sinks in the middle.
6. Complete the following table as a summary of the specific reactions proteins undergo during the preparation, storage or presentation of food items.

FUNCTIONAL PROPERTY OF PROTEIN	OCCURS WHEN...	EFFECT ON FOOD (FUNCTION)
Denaturation		Irreversible; may be visible (eggwhites) or invisible (marinaded meats)
Coagulation		
Aeration		
Gelation	Globular proteins unravel and bind liquids, forming soft semi-solids.	
Gluten formation		
Maillard reaction	Proteins are exposed to dry heat in the presence of carbohydrates.	Browning and crisping, with slight caramelisation, also providing flavour change
Myoglobin browning		
Oxidative browning		Dull brown colour, stronger flavour

Do an activity

7. Cappuccinos are popular beverages that rely on the functional properties of protein to create the product. In groups, discuss whether the creation of the foam is a demonstration of denaturation or coagulation? Using the specific terminology relating to this topic, outline the reasons for your opinion.

Carbohydrates

Now that we have learned about proteins, let's look at the functional properties of another of the main nutrients in foods: **carbohydrates**. You will remember that there are three types of carbohydrate found in foods: polysaccharides, disaccharides and monosaccharides. Polysaccharides include starches, pectin and dietary fibre, while mono- and disaccharides are sugars. You would recognise that foods containing a lot of sugar taste different from those made up mainly of starch — but taste is not the only difference the carbohydrates provide in the food. Polysaccharides undergo different reactions from sugars during preparation and cooking. It is therefore necessary to discuss the functional properties of polysaccharides (starch and pectin) separately from those of sugars (mono- and disaccharides).

Functional properties of starch

Starch is a common polysaccharide found in grains such as rice and wheat and the products made from them, and in vegetables such as potatoes and cassava (tapioca). The size of the starch granules found in these foods can vary, as can their specific components. Starch molecules are made up of two substances: amylose and amylopectin. The ratio of amylose to amylopectin varies and this creates different qualities in starches from different sources. Most starches contain around 25 per cent amylose, but some selectively bred varieties of corn contain more than 50 per cent. When starches with a higher proportion of amylose are used in food preparation, they produce foods with different qualities from those with higher proportions of amylopectin. For example, starches containing more amylose are used to produce semi-solid, opaque gels, whereas those with more amylopectin produce clearer gels that are more a thick liquid ('viscous') than solid. High-amylose cornstarch and rice starch are used to make edible packaging films, such as those used on some lollies and sweets.

Gelatinisation of starch

Starch can thicken mixtures by absorbing liquid in the presence of heat. This process is called **gelatinisation** and takes place regardless of the type of starch. The physical qualities of the gel produced, however, are determined by the ratio of amylose to amylopectin and the method used to prepare the food.

Plain wheat flour, cornflour and arrowroot are the most common ingredients used to thicken liquids (thickening agents) in a domestic kitchen. Wheat flour contains a high proportion of amylose and produces thick but very cloudy gels. However, plain flour also contains 10 to 15 per cent gluten, so it has less starch per tablespoon than the other thickening agents. Cornflour contains a higher proportion of amylose than arrowroot so it produces a thicker and cloudier sauce than arrowroot. In the hospitality industry, potato starch is often used to give soups a slightly thicker consistency; because it has a higher proportion of amylopectin, it does not form lumps or a gel. Food manufacturers also use a variety of modified starches to produce the specific consistencies they desire in their products.

Factors that affect the gelatinisation of starch

Like proteins, the functional properties of carbohydrates are affected by changes in temperature, pH, agitation and the presence of other ingredients. Let's look at how each of these factors affects the way starch reacts during food preparation and cooking.

Temperature

Gelatinisation requires warm, moist conditions. Starch can not thicken cool liquids as it is not soluble in cold water. In order for starch to absorb liquid, the outside of the starch granule needs to be softened by heating. At 64–72 °C, the outer shell of the starch granule softens, allowing it to gradually swell up with water. When the temperature reaches 86–90 °C, the swollen starch granules burst and the amylose and amylopectin they contain are released, capturing the liquid and causing it to thicken and form a gel. Gelatinised mixtures should be boiled for at least 1 minute to ensure that all the starch has reached the desired temperature; any ungelatinised starch will give the finished product a 'pasty taste'. If all the starch has fully gelatinised, the sauce should look smooth and shiny.

When a gelatinised mixture cools, it forms a more rigid gel. Some cold gels, such as custard made from custard powder, rupture and 'weep' liquid when cut or stirred. Also, gels break when reheated; that is why lumps appear when flour-thickened gravy is reheated.

A **roux** made from wheat flour is often used in the preparation of sauces and gravies. When making a roux, heat equal quantities of melted fat and flour in a pan for 1 to 2 minutes to soften the outer shell of the starch granules. (Such heating of starch in the absence of water produces **dextrins**, as discussed on page 94.) Remove the roux from the heat and allow it to cool slightly before adding cold liquid; stir the mixture constantly while adding the liquid to prevent lumps forming.

Agitation

Agitation is essential for the creation of a smooth, gelatinised sauce. Gelatinisation takes place within a specific temperature range in the presence of liquids; if only some granules of starch are exposed to the liquid at the right temperature, they burst and trap the liquid around them, preventing it from reaching other starch granules. This results in visible lumps of starch throughout the mixture.

To prevent lumps forming in a sauce thickened with starch:

- mix flour with cold water in a cup rather than adding dry flour to a warm mixture, such as when thickening a stew.

Gradually adding dry flour to the cold liquid, rather than adding liquid to the flour, minimises the risk of lumps forming. The cold water–flour mixture should be stirred immediately before adding it to the warm liquid, because starch is insoluble in cold water and settles to the bottom of the cup. This cold starch mixture should be added slowly to the warm liquid while stirring.

- add only cold liquid to a starch mixture. Adding a hot liquid causes some starch molecules to gelatinise on contact with the liquid, forming lumps.
- cool roux mixtures before adding cold liquids so that none of the starch molecules is at a temperature over 86 °C when the liquid is added. The mixture should be stirred while the liquid is added so that all starch molecules have equal access to the liquid.
- stir roux mixtures constantly from the time the liquid is added until the sauce boils. If the mixture is not stirred, the starch molecules closest to the bottom of the pan gelatinise before those at the top, which are not as hot. Stirring the mixture disperses lumps throughout the mixture. Whisking the sauce may help remove the lumps and improve the texture of the sauce.

The impact of other ingredients and pH

The gelatinisation process can be affected by several additives, particularly sugar and acid. Sugar competes with the starch for water, so not as much water is available for the starch to absorb. Also, the sugar raises the temperature at which the starch granules start gelatinising. This makes the mixture more liquid and less likely to remain stable.

- Acids such as those found in vinegar, lemon juice and other fruit juices have an even greater effect on the thickness of the final mixture. In the presence of heat, acids break down the large starch molecules into smaller molecules of dextrin, which have much less thickening power than starch. When preparing a starch-thickened mixture that requires the addition of an acid (a lemon pie filling, for example), add the acid to the starch paste *after* gelatinisation has occurred. In other words, add the lemon juice after the mixture has thickened and boiled.
- Equal quantities of fat and flour in a roux ensures that there is adequate fat to disperse the starch and minimise the risk of lumps forming.
- The addition of sugar to a mixture, such as custard, results in a softer, less stable gel.

Dextrinisation of starch — 'browning reaction'

Dextrinisation is defined as the process in which starch breaks down into dextrins. One of the most common examples occurs when bread is toasted.

The polysaccharide starch can undergo a partial chemical breakdown into shorter glucose chains called dextrins. Dextrins are the halfway point in the complete breakdown of a starch molecule into the monosaccharide glucose.

starch (polysaccharide) → dextrins → maltose (disaccharide) → glucose (monosaccharide)

The chemical structure of dextrins is different from that of starch, giving dextrins several characteristics that are quite different from those of starch.

1. Starch does not dissolve in cold water but dextrins do.
2. Starch has a floury taste, and dextrins are sweet.
3. Starch gelatinises, but dextrins are poor thickening agents.

Dextrinisation can be caused by:

- exposure of the starch granules to dry heat (the most common method)
- the action of acids
- the action of certain enzymes.

The dextrinisation process is the reason for:

- bread turning brown during toasting
- toasted commercial breakfast cereals being brown and having a sweeter taste than raw cereals
- the brown crust forming when bread or pastries are baked
- gravies being brown
- the brown crust forming on baked starchy vegetables such as potatoes and carrots.

Dextrins are used commercially as both a colouring agent and a sweetening agent.

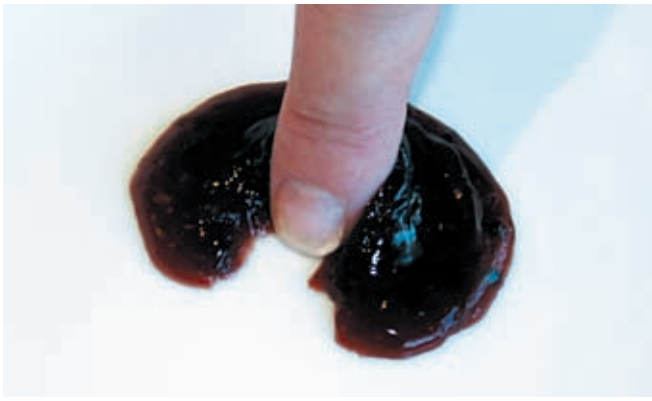
Functional property of pectin

Pectin is a polysaccharide found in the liquid extract of many fleshy fruits. It is used to create fruit gels such as jams and jellies.

Fruits with a high pectin content include lemons, limes, cumquats, apples, blackberries, cranberries, currants, grapes, guavas and sour plums. Fruits with a moderate pectin content include oranges and quinces. The more overripe a fruit is, the less pectin it contains.

When making fruit pectin gels, cut the fresh ripe fruit into small pieces. If the fruit has a moderate or low pectin content, add acid or use a commercially produced pectin concentrate. Simmer the fruit, so the softened fruit releases its pectin. Unlike starch, pectin dissolves easily in water, so no lumps form during the thickening process. Add sugar and boil the mixture rapidly to evaporate some of the water and concentrate the sugar until the setting point is reached.

To determine the point at which the pectin concentration is high enough to form a gel on cooling, two methods are used. By the time the mixture reaches 103–105 °C (when tested with a cooking thermometer), the jam or jelly should be ready. To make sure, scoop a small amount of the mixture onto a cold metal spoon and allow to cool a little. Tilt the spoon slightly and allow the mixture to drip. The jam or jelly is ready when the mixture drips in heavy, separate drops.



Pectin is responsible for the gelled qualities of jam. Acid must be present for pectin to gel. Test the level of gelling by placing a small quantity of jam on a saucer and cooling it slightly. If the surface wrinkles and leaves a path when a finger is dragged through it, the jam is ready for bottling.

If the jelly fails to give a positive test at 105 °C, add a small amount of lemon juice. Care needs to be taken to avoid overcooking, which destroys the pectin and prevents the product from thickening. The gel may also develop an undesirable colour and flavour when overcooked.

REVIEW QUESTIONS

Remember

1. Outline the physical and chemical differences between starches, dextrins and sugars.
2. Describe what pectin is and identify three sources of pectin.

Apply

3. Suggest reasons why cornflour is commonly used as a thickening agent when preparing sauces and stews.
4. **a)** Define the term 'roux' and give examples of its use in food preparation.
   b) Consider the process used to make a roux.
      i. Different types of roux are described by their colours: white, blonde and brown. How is the process of making the roux altered to achieve these different colours?
      ii. Does the colour or type of roux affect other characteristics of the sauce? Explain your answer.

Do an activity

5. Do you think you can thicken stews with pectin or use starch to make jam? In pairs, compare and contrast the gelatinisation of starch and gelation of pectin. Consider the carbohydrates used, the methods of preparation and the characteristics of the end result.

Functional properties of other carbohydrates: sugars

Crystallisation of disaccharides

The three kinds of disaccharides are sucrose, lactose and maltose. By far the most common type of disaccharide used in making foods is sucrose, (common table sugar). Disaccharides are crystalline in structure, sweet to taste, and dissolve easily in water. Monosaccharides (glucose, fructose and galactose) have similar characteristics but take a long time to produce very small crystals.

Crystallisation is a functional property of sugar related to its ability to dissolve and reform crystals. This property is used in the making of confectionery such as toffee, brittle, fudge and caramel. The sugar is dissolved in a liquid and heated, during which time the liquid evaporates, concentrating (supersaturating) the sugar solution. As the solution cools, the sugar re-forms into crystals. If the solution is not stirred and cools quickly, the crystals will be very small and the texture of the product smooth. Movement and slow cooling encourage the development of larger crystals, altering the texture of the product.

Factors affecting the crystallisation of disaccharides

As with proteins and starch, the functional properties of sugar are influenced by temperature, pH, agitation and the presence of other ingredients. These factors each have a specific effect on qualities of foods that rely on sugar crystallisation.

Temperature

The hotter the water is, the greater the amount of sucrose that can dissolve in it. For example, water at 40 °C dissolves 238 grams of sugar, while water at 115 °C dissolves 669 grams. Also, if there is more sucrose in a solution, more crystals can form. When more crystals form, the product is more solid. The product's solidity is also determined by the length of time the solution is exposed to heat. As a solution boils, liquid evaporates, increasing the concentration of sugar (becoming supersaturated) and the temperature of the solution. Extremely high temperatures (above 150 °C) produce brittle crystalline products that also show signs of caramelisation.

TABLE 5.3 The heating of sugar

TEMPERATURE	TERM	FOOD PRODUCT
106–113 °C	Thread	Cake decoration
112–116 °C	Soft ball	Fudge, fondant
118–121 °C	Firm ball	Toffee, soft caramels
121–130 °C	Hard ball	Hard toffees and caramels
132–143 °C	Soft crack	Butterscotch, nougat
149–154 °C	Hard crack	Barley sugar
160–177 °C	Caramel	Nut brittle, praline

Sugar thermometers are used to accurately measure the temperature of sugar solutions and help control the solidity of confectionery products. Consistency can also be measured by dropping small quantities of sugar solution in cold water. The colder the water is, the faster the sugar solidifies (crystallises) and the more accurately the texture of the finished product can be managed. Be extremely careful with molten sugar as the temperatures reached are dangerous. What would you do if some molten sugar came in contact with your skin?

Acidity

Disaccharides are formed when two monosaccharides chemically join to create a new sugar molecule. Monosaccharides can be purchased from the supermarket in the form of liquid glucose and honey, which contains glucose and fructose. They can also be made during the crystallisation process by adding an acid to the liquid, dissolving the sugar. The acid, commonly vinegar or cream of tartar, splits the bond joining the disaccharide, creating separate monosaccharides, referred to as 'invert sugars'. Monosaccharides crystallise less readily than disaccharides and so inhibit the development of large crystals in a solution; this gives the confectionery a smoother texture. If present in large quantities, however, the product remains runny rather than solidifying.

Agitation

Stirring sugar solutions encourages crystals to develop. If a solution is stirred while hot, crystallisation starts too soon, giving the product a grainy texture. It is, therefore, essential that sugar is stirred into a liquid and fully dissolved before the mixture boils. Stirring after boiling point is reached encourages crystals to form.

Some products, such as fudge, require beating to develop their characteristic textures. Waiting until the solution has cooled to at least 40 °C before stirring or beating encourages small crystals to form, giving the product a smooth texture. Beating should continue until crystallisation is complete and the product becomes solid. When fudge is made commercially, the beating is done by machines and the final texture is usually much better than can be achieved with hand beating.

Other products, such as caramel, require beating while heating. Caramel is made by adding other ingredients, such as milk products and butter. The butter helps to minimise the size of the crystals (the reason is discussed later in this section) and the milk product contributes to the flavour of the caramel. Because the combination of ingredients results in a thick mixture, it must be stirred throughout the cooking process to ensure that the mixture heats evenly and does not burn at the base of the pan.

Presence of other ingredients

Additives can have a dramatic effect on the crystallisation process.

- Sugar in crystal form attracts molten sugar, encouraging the development of large crystals and giving the product an unappealing grainy texture that is similar to candied honey.
- Monosaccharides, added as glucose, honey or created by adding acidic ingredients, are often included to ensure a smooth consistency in the final product. As previously discussed, monosaccharides prevent the development of large crystals and, if added in excess, prevent all crystal development, resulting in a liquefied rather than crystalline structure.
- Fat (such as butter or cream) surrounds small sugar crystals as soon as they form, preventing large crystals from forming. This ensures the smooth texture of, for example, butterscotch and caramel.

Sucrose melts at 186 °C. Above that temperature, acids form and the molten sugar turns a light brown, or caramelises. If the heat is not removed, the liquid sugar

EXPERIMENT

Caramelisation of sugar

Syllabus outcome

Students learn to:

- investigate through experimentation the factors that affect the functional properties of foods.

Contributes to outcome:

- applies an understanding of the sensory characteristics and functional properties of food to the preparation of food products.

Aim

To caramelise sugar to produce barley sugar and violet crumble

Equipment

2 cups white sugar
2 small heavy saucepans
2 metal oven trays
measuring spoons
pinch bicarbonate of soda

Method

1. Pour the sugar into the saucepan.
2. Put saucepan on low heat and shake the pan gently so the sugar does not stick to the bottom of the pan. Continue shaking until all the sugar has melted.
3. After the sugar has melted, leave the pan undisturbed on the heat.
4. Continue heating until the molten sugar turns a golden brown.
5. Pour the liquid onto a metal oven tray to cool. Leave undisturbed until completely solidified. Observe what happens as the liquid solidifies. Record your observations in an appropriate form.
6. Repeat steps 1–5, but add a pinch of bicarbonate of soda to the molten sugar just before pouring the liquid onto the metal tray.

Note: Place water into saucepan after sugar is removed and return to heat. (This dissolves the remaining sugar and makes the saucepan easier to clean.)

Conclusions

Which confectionery product has:

a) the greatest volume
b) the darkest colour
c) the lightest texture
d) the most lustrous surface?

becomes darker and darker until it is black. The breakdown of sucrose occurs in this way:

sucrose + heat → glucose + fructose + water →

acids + glucose

This process occurs *without* water. Water may have originally been present (as when making confectionery products) but it evaporates before caramelisation takes place.

Caramelised sugar adds a distinctive flavour to foods. You will often see caramel flavouring listed as an ingredient of commercially produced flavoured milks, cakes, breakfast cereals and ice-creams. 'Burnt sugar' is a variety of frosting that involves caramelising sugar and adding it to butter.

Summary

Table 5.4 summarises the reactions of carbohydrates and the function of each reaction in food preparation, storage and presentation.

TABLE 5.4 Carbohydrate reactions

CARBO-HYDRATE	FUNCTIONAL PROPERTY	FUNCTION IN FOOD
Starch	Gelatinisation	Thickening agent
Starch	Gelation	Formation of gels
Starch	Dextrinisation	Browning and sweetening
Pectin	Gelation	Formation of fruit gels
Sucrose	Crystallisation	Making of confectionery
Sucrose	Caramelisation	Browning and flavouring

Lipids

Lipids include all fats and oils. The two important reactions that lipids undergo during preparation are emulsification and aeration.

Functional properties of lipids

Emulsification

Lipids do not naturally mix with water. They can be shaken to form a temporary mixture, as when you make an oil and vinegar salad dressing, but, after a short time, the oil floats to the surface of the vinegar.

An **emulsion** is a stable combination of oil and water. Emulsification is the process that produces an emulsion. To form a permanent emulsion, the following techniques are used.

- The fat globules are reduced in size to a small uniform diameter and evenly distributed throughout the water. This process is called **homogenisation** and is used in the production of regular and low-fat milk. Skim milk has no fat present, so homogenisation is not needed.

Homogenisation is an emulsifying process that distributes lipids evenly throughout a liquid, improving the product's flavour, texture and saleability. Would you buy a food product with a layer of oil sitting on top of it?

REVIEW QUESTIONS

Remember

1. Write a definition for each of the functional properties of carbohydrates in your own words. Include the type of carbohydrate that displays the property, and give two food examples that use the property.
2. Copy and complete the table below in your workbook to compare the effects of different factors on the functional properties of carbohydrates.

Apply

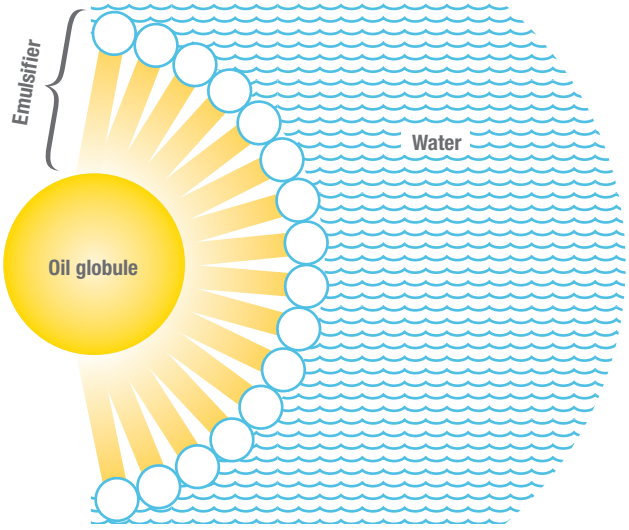
3. Identify the functional properties relating to the browning of carbohydrates. For each property, list the characteristics that these reactions develop in the food.
4. Crystallisation and caramelisation are both used in making confectionery.
   a) Outline the differences between them.
   b) Explain whether it is possible for one to occur without the other.

Do an activity

5. Find a recipe for making caramel.
   a) Identify factors in the recipe that encourage the development of larger sugar crystals.
   b) Identify factors that encourage the development of small sugar crystals.
   c) Explain why a darker colour affects the hardness of the product.
   d) Make the recipe, heating the mixture long enough to produce a firm, chewy caramel.

FACTOR	DEXTRINISATION	GELATINISATION	GELATION	CRYSTALLISATION	CARAMELISATION
Temperature					
pH					
Agitation					
Presence of other ingredients					

- A special ingredient is added to stop the separation of lipid and water. These agents are called emulsifiers or emulsifying agents. An **emulsifier** is a substance that has an attraction for both the lipid and the water, acting as a bridge.



One end of an emulsifier is attracted to oil, while the other is attracted to water. Emulsifiers allow oils to be dispersed within foods, providing foods with the necessary characteristics without appearing oily. Would consumers make healthier food choices if they could see more of the fat present in their foods?

Emulsions, although stable, can be destroyed under certain conditions. High temperatures, freezing, agitation after formation and the addition of salt break down an emulsion. If the emulsifier bridge breaks, the lipid and water separate.

Factors affecting the emulsification of lipids

The functional property of emulsification is affected by:

- *temperature*
  - High temperatures cause lipids to become more fluid and water to move more rapidly, making it harder for the emulsifier to hold them together.
  - Freezing changes the state of both water and oil; crystallisation causes the connections with the emulsifying agent to break. When the product defrosts, the oil and emulsifier float to the top.
- *agitation*
  - When adding oil for emulsification, the mixture must be beaten rapidly so that the oil has access to the emulsifier. The general rule is to beat the mixture while adding oil slowly, and to add no more oil at any time than the amount already in the mixture.
  - Beating or continuous jarring of an emulsified mixture can break the connections between the emulsifier and the oil, destroying the emulsion.
- *addition of salt*
  - The chemical attraction between water and salt is greater than the chemical attraction between water and the emulsifier. Therefore, adding salt to an emulsion causes water in the emulsion to separate and move towards the salt.

Emulsifiers can be found naturally in foods. Examples of natural emulsifying agents are lecithin in egg yolk, and gelatin. Synthetic emulsifiers such as glyceryl monostearate, SPANS and TWEENS can be manufactured in a laboratory.

Parts of an emulsion

An emulsion has three parts: the dispersed phase, the continuous phase and the emulsifier. The make-up of margarine demonstrates these three parts. Margarine contains:

- no more than 20 per cent water, called the dispersed phase because it is the substance present in the smallest quantity, and it is suspended as droplets
- no more than 80 per cent lipid content, called the continuous phase because it is the substance in the largest proportion in the emulsion
- emulsifier(s), which are able to mix with both the lipid and water components.

Types of emulsions

There are basically two kinds of emulsions. One has more lipid than water — 'water in oil' emulsion — and is found in butter and margarine. The other has more water than lipid — 'oil in water' emulsion — and is found in mayonnaise, salad dressings, gravies, cream soups and pie fillings.

Aeration of fats

Aeration occurs when air is added to a mixture, affecting its consistency and texture. The plasticity of saturated fats makes them more effective at aerating mixtures than unsaturated oils. Here are some examples of how fat is used to aerate products:

- creaming butter and sugar together when preparing cake batters — the texture of the mixture becomes light and fluffy, and the colour of the fat changes from a solid yellow to a light cream colour. This occurs because, as the sugar crystals are pressed into the fat, a small quantity of air is included in each indentation.
- 'rubbing in' (discussed on page 91) — this process also introduces air into a mixture
- layering fat between layers of gluten in the making of puff pastry
- whipping cream to form a foam.

Foams

Foams are formed when air is whipped or beaten into a liquid. Whipped cream is an example of a foam in which a film of fat and protein surrounds the air bubbles. The whipping action partly solidifies the fat and partly denatures the protein. After these two changes occur, the liquid becomes a fairly stable solid.

The fat content of cream needs to be 22 to 35 per cent before it can be whipped. Evaporated milk has a higher concentration of protein and fat than that of ordinary milk,

as a result of having 60 per cent of its water removed, and therefore it can be whipped if thoroughly chilled first.

Factors that affect the foaming of cream are:

- *the age of the cream* — Cream whips better if it is 2 to 3 days old.
- *temperature* — At 1–7 °C, the cream has a thicker, less runny consistency and whips more easily.
- *amount of beating* — When overbeaten, the fat globules clump on the surface of the air bubbles and the whipped cream takes on a curdled appearance.
- *addition of gelatin* — Gelatin makes the cream less runny so a thicker foam is produced. Thickened cream bought in shops contains 1 per cent gelatin, as well as alginate or another substance that makes the cream foam more quickly and that prevents overbeating.
- *addition of sugar* — This decreases the volume of the foam but, if added towards the end of beating, decreases the likelihood of the cream curdling and gives the cream a more lustrous appearance. Icing sugar is generally used.

When oxygen reacts with food

Another significant reaction is oxidation. Oxidative reactions affect the quality of the food.

- Some proteins undergo oxidative reactions. For example, the surface of red meat turns brown when exposed to air for a long time (see page 91).
- Lipids undergo 'oxidative rancidity' if exposed to air for a long time, especially in high temperatures. Oxygen reacts at double bonds along the carbon chain of a fatty acid; this means that oils are more likely to go **rancid** than fats. The reaction with oxygen causes the lipid to spoil, creating odours, a bitter taste and a lower smoking point. Oxidative reactions in oils, and products containing them, may be prevented by adding an **antioxidant** (see page 185 for examples). Antioxidants are substances that react readily with oxygen, delaying oxidative reactions with the food.
- Carbohydrates are not affected by oxidative reactions.
- Fruit and vegetables are susceptible to oxidative reactions because, when cut, their enzymes are exposed and react easily with air, causing the surface of the food to brown. Sometimes this reaction is desirable, such as in the making of cider, but more often it is not. Enzymic oxidation can be reduced or prevented by:
  - packaging the food to eliminate as much air as possible
  - coating food with an acid, such as lemon juice, to denature the enzymes
  - coating the food with an antioxidant, such as citrus juices (containing the highly reactive ascorbic acid) and pineapple juice (containing a sulfur compound and sugar solution).

REVIEW QUESTIONS

Remember

1. Define the term 'emulsion' and give three examples of food products that are stable emulsions.
2. Most milk in cartons sold in Australia has been 'homogenised'. What does this term mean and what benefit, if any, does it provide consumers?
3. Explain, using examples, how emulsifying agents work.

Apply

4. Many different types of cream are available to consumers, including fresh cream, thickened cream, light cream, double cream and sour cream.
   a) Explain the difference in characteristics between these products.
   b) Which of these products does not aerate or foam when beaten? Explain why.

Do an activity

5. Earlier in the chapter we discussed how gluten strands are created. Predict what differences would occur in the flour (gluten) if you were to
   a) place some flour in a loosely woven fabric and hold it under cold running water until the water ran clear
   b) rub some butter into the flour before placing it into the fabric, then hold it under cold running water.

   Conduct a comparative experiment to see if your predictions were correct.

CHAPTER 5 SUMMARY



- The functional properties of nutrients determine the physical characteristics of foods.
- The physical characteristics created in the food are influenced by specific factors such as temperature, pH, agitation and the presence of oxygen, enzymes and other ingredients (such as sugar, salt, fat and water).
- Under certain conditions, proteins denature, coagulate, form gels and foams, and become brown in colour (via the Maillard reaction, oxidation and non-enzymic reactions.
- The shape of a protein influences the speed at which it denatures.
- Because of their significant structural differences, different types of carbohydrates have different functional properties in foods. Pectins form gels, starches dextrinise, gelatinise and form gels, while sugars crystallise and caramelise.
- The size and type of a carbohydrate chain determines the textural qualities of the final product.
- Lipids can aerate and be emulsified, thereby significantly affecting the physical (especially textural) qualities of the foods that contain them.
- Plastic fats, such as solid butter and cream, are more effective at providing aeration than oils.
- Enzymes that occur naturally in foods can react with oxygen, causing enzymic browning. This type of browning reaction can be prevented by excluding oxygen, adding substances that react with air and minimise similar reactions in the food, and temperatures above 60 °C, which denature the enzyme so that it cannot function effectively.

KEY TERMS

aeration
ageing process
amino acids
antioxidant
bacteria
bacterial culture
carbohydrate
catalyst
coagulation
crystallisation
denaturation
dextrin
dextrinisation
emulsifier
emulsion
enzyme
enzymic browning
foam
gel
gelatinisation
gelation
gluten
helix
homogenisation
lipids
Maillard reaction
pectin
peptide bond
pH
protein
rancid
roux
starch